1. Introduction and Project Objectives

Air Quality Analytics and Pollution Hotspot Detection in India

Study question
Which monitoring stations repeatedly experience high pollution, and how much observed data supports the result?

- Analyze historical CPCB observations for the complete year 2024.
- Select a pollutant, state, city and period in an interactive dashboard.
- Identify persistent hotspots and inspect monthly patterns.
- Execute the SQL in BigQuery and validate the outputs independently.

Project contribution

Persistence screening with visible coverage, pollutant-specific averaging rules and explicit missing-data limitations.

2. Dataset and Big Data Scope

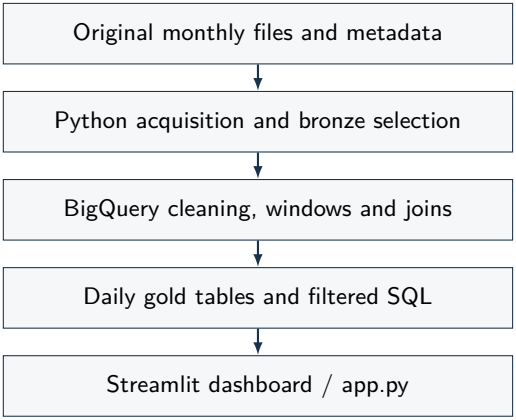
Verified measure	Count
Monthly source files	12
All acquired source measurements	46,937,170
Selected CPCB measurements	27,021,875
Stations in the selected subset	533
Eligible pollutant-station-days	1,115,522
Combined station-days	165,652

- One record means one pollutant at one station and time.
- Seven pollutants: PM2.5, PM10, NO2, SO2, O3, CO and NH3.
- Source: XKDR Forum's CPCB compilation, CC BY 4.0 [1].

Why BigQuery helps

Cloud SQL processes the measurements and returns filtered summaries. This demonstrates warehouse analytics without claiming the dataset cannot run on a laptop.

3. System Design and Data Lifecycle



Quality and provenance
Checksums and row reconciliation. IST dates with unique UTC keys. No missing-value filling.

Independent check
DuckDB reference outputs verify cloud results. It is a validation path.

Application access
Read-only viewer credentials, bound parameters and a ten-minute cache.

4. Pollutant Rules and Combined Pollution Level

Pollutant	Reference	Daily comparison
PM2.5 / PM10	60 / 100 $\mu g/m^3$	24-hour mean
NO2 / SO2	80 / 80 $\mu g/m^3$	24-hour mean
NH3	400 $\mu g/m^3$	24-hour mean
O3	100 $\mu g/m^3$	Highest eligible 8-hour mean
CO	2 mg/m^3	Highest eligible 8-hour mean

- Each day needs at least 16 valid hourly observations.
- An 8-hour window needs six readings and stays within the IST day.
- Combined level: worst available CPCB band, from 0 Good to 5 Severe.
- Require at least three valid pollutants, including PM2.5 or PM10 [2].

Interpretation

The combined measure is an ordinal screening level, not official numeric AQI. Missing pollutants can hide a worse band. O3 upper bands use the source table's one-hour convention.

5. Coverage and Persistent Hotspot Detection

Coverage

$$100 \times \frac{\text{valid observed days}}{\text{calendar days in period}}$$

High-day fraction

$$100 \times \frac{\text{high days}}{\text{valid observed days}}$$

Missing days never count as clean days.

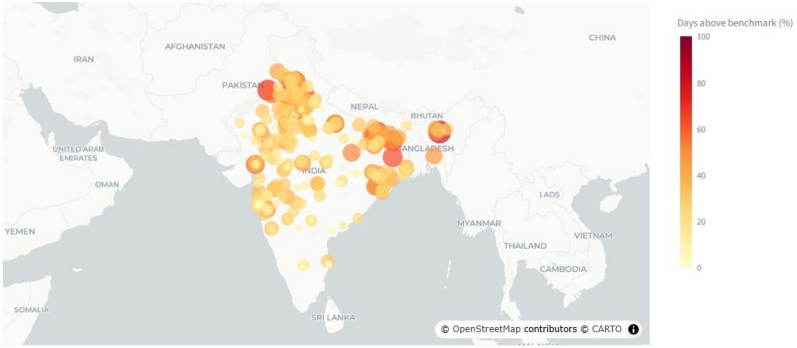
Default persistence rule

- Overall day coverage at least 75%.
- Each counted month also has at least 75% coverage.
- At least 10% high days in each counted month.
- At least three high-pollution months.

Transparent project choices

Thresholds are adjustable. PM2.5 persistent counts change from 370 at 60% coverage to 342 at 75% and 250 at 90%. This changes eligibility, not the measured air quality.

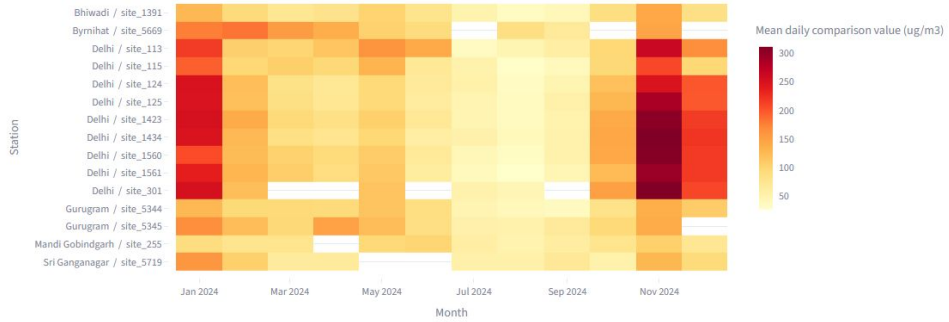
6. Dashboard Output: Pollution Hotspots



11 eligible stations have no source coordinates and remain in the ranking. The online basemap depicts monitored points, not interpolated pollution across India.

- PM2.5 default view: 416 coverage-eligible stations, 342 persistent.
- Sector-51, Gurugram: 84.2% high days from 330 valid days.
- Eleven eligible stations lack coordinates and remain in the ranking.

7. Dashboard Output: Monthly Patterns

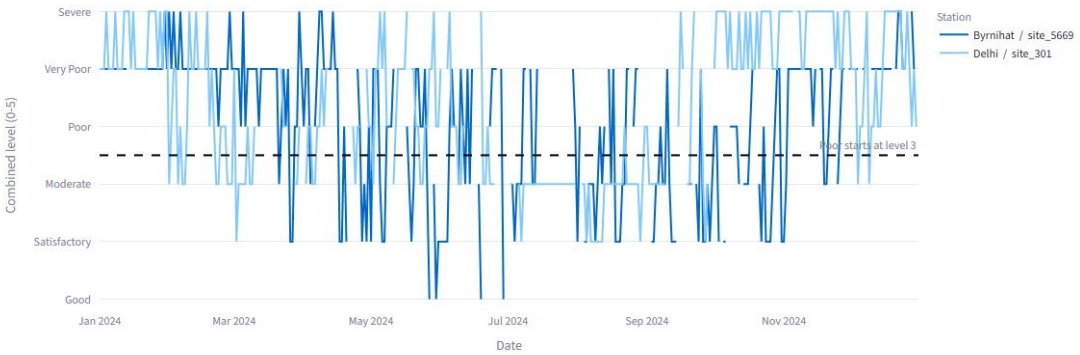


Blank cells lack sufficient day coverage. A monthly concentration summary or poor-day fraction is descriptive, not an official monthly AQI.

- Means of daily PM2.5 values for the top 15 ranked stations.
- Blank cells fall below the coverage rule. A second heatmap shows coverage.
- One year describes monthly differences, not recurring seasonality or causes.

8. Dashboard Output: Combined Level and Comparison

432 coverage-eligible stations and 222 persistent hotspots



- Derived combined level from 0 Good to 5 Severe. Poor starts at level 3.
- Daily details show available pollutants. Compare up to three stations and export CSV.

9. Results, Validation and Course Alignment

Default annual screening

View	Eligible	Persistent
PM2.5	416	342
PM10	412	372
NO2	425	35
SO2	432	4
O3	421	167
CO	435	172
NH3	389	0

Validation evidence

- 32 tests passed after cleanup.
- Four cloud output tables match the local reference.
- Three dashboard aggregation scenarios match.

ED547 connection

Modules IV and V: ETL, Google BigQuery, SQL and Python. Modules I and II: lifecycle, data quality and warehouse storage.

Verified execution

Retained job IDs and processed-byte records establish real BigQuery execution. Hadoop, Spark and Airflow are not implemented. No controlled speed benchmark is claimed.

10. Conclusion, Limitations and References

Conclusion

The project links large-scale cloud processing to station-level evidence, coverage-aware persistence and reproducible outputs.

Limitations

- Historical 2024 data only. No prediction or attribution of causes.
- Uneven monitoring, missing pollutants and 53 missing-coordinate stations.
- Derived combined level and adjustable project thresholds.
- Sandbox tables expire after 60 days. Internet and credentials are required [3].

References

[1] XKDR Forum. India Air Quality Database. `https://airquality.xkdr.org/`
[2] Indian Economic Service. National Air Quality Index.
`https://ies.gov.in/arthapedia/concept/national-air-quality-index`
[3] Google Cloud. BigQuery sandbox. `https://docs.cloud.google.com/bigquery/docs/sandbox`
Accessed 1 November 2026. Results: retained project audits and analytical exports.